

LIU WENYANG

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PROFESSIONAL SUMMARY

Computer Vision researcher focused on **image/video restoration, super-resolution, degradation modeling, and efficient vision models for real-world imaging and sensing systems**. Experienced in deep learning-based image enhancement, corrupted media recovery, and deployable AI models, with research interests in visual quality improvement, computational imaging, and mobile/edge visual intelligence.

Highlights: 1 first-author CVPR paper | 2 first-author IEEE TMM papers | 7 first/co-first-author publications | 20 publications, 300+ citations, h-index 8, i10-index 7

Research Focus: Image/Video Restoration | Super-Resolution | Degradation Modeling | Image Quality Enhancement | Efficient Vision Models | Multimodal Visual Understanding

EDUCATION

Nanyang Technological University, Singapore

Jul 2022 – Nov 2025

Ph.D. in Electrical and Electronic Engineering

- Thesis: *Visual Image Restoration and Super-Resolution Using Deep Learning*
- Supervisors: Prof. Chau Lap-Pui, IEEE Fellow, and Assoc. Prof. Yap Kim-Hui

Chongqing University, China

Sep 2013 – Jul 2020

B.Sc. in Internet of Things | M.Sc. in Computer Science

- Research focus: efficient computing systems, low-power video decoding, and AI acceleration.

TECHNICAL SKILLS

- **Imaging & Vision:** Image/Video Restoration, Super-Resolution, Denoising, Image Enhancement, Degradation Modeling, Visual Representation Learning
- **Models:** CNNs, Transformers, Diffusion Models, Vision–Language Models, CLIP, LLaVA, Qwen-VL, Knowledge Distillation, Continual Learning
- **Tools:** PyTorch, TensorFlow, HuggingFace, OpenCV, NumPy, Pandas, Linux, Git, Python, C/C++

WORKING EXPERIENCE

Research Fellow, Hyundai–NTU–A*STAR Corporate Lab, NTU

Nov 2025 – Present

- Conduct research on robust visual perception for smart manufacturing scenarios involving cluttered layouts, occluded objects, small parts, and domain variation.
- Develop open-world object detection and segmentation methods by combining language prompts and visual references to improve object localization and mask quality in industrial scenes.

Research Associate, School of Electrical and Electronic Engineering, NTU

Jan 2021 – Oct 2025

- Led research on image/video restoration, lightweight super-resolution, and corrupted multimedia recovery, resulting in publications in CVPR, IEEE TMM, and JVCIR.
- Developed deep learning models for visual degradation scenarios, including bitstream corruption, watermark noise, low-resolution inputs, and incomplete multimedia files.
- Built visual and byte-level representation models for multimedia understanding, file fragment classification, and corrupted visual data analysis.

SELECTED RESEARCH EXPERIENCE

Image and Video Restoration under Corrupted Bitstreams

- Developed compensation, alignment, and reconstruction strategies to recover damaged visual content from corrupted JPEG images and bitstream-damaged videos.
- Papers: *CVPR 2023* on corrupted JPEG image restoration and *NeurIPS 2023* on corrupted video recovery.

Lightweight Image Super-Resolution and Enhancement

- Explored prompt-guided restoration, adaptive token aggregation, and degradation-aware representation learning for efficient image quality improvement.

- Papers: *IEEE TMM 2026* on lightweight super-resolution, *JVCIR 2025* on watermark removal, and *IEEE TMM* under review on adaptive super-resolution.

Visual and Multimodal Representation Learning

- Designed byte-sequence, byte-image, and multimodal representations for file fragment classification, anomaly detection, open-vocabulary perception, and video understanding.
- Papers: *IEEE TMM 2024* on multimedia file fragment classification, *ACM MM 2025* on open-vocabulary vision, *KBS 2025* on CL-HOI, and *ACM MM* under review on industrial anomaly detection.

PUBLICATIONS

- [1] **W. Liu**, Y. Wang, K.-H. Yap, L.-P. Chau. “Bitstream-Corrupted JPEG Images Are Restorable: Two-Stage Compensation and Alignment Framework for Image Restoration.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. [CCF-A] [\[Image Restoration\]](#)
- [2] **W. Liu**, C. Cai, J. Gao, K. Wu, Y. Wang, K.-H. Yap, L.-P. Chau. “PromptSR: Cascade Prompting for Lightweight Image Super-Resolution.” *IEEE Transactions on Multimedia (TMM)*, 2026. [CCF-A, JCR Q1] [\[Image Restoration\]](#)
- [3] **W. Liu**, J. Gao, K.-H. Yap. “SSH-Net: A Self-Supervised and Hybrid Network for Noisy Image Watermark Removal.” *Journal of Visual Communication and Image Representation (JVCIR)*, 2025. [CCF-C, JCR-Q2] [\[Image Restoration\]](#)
- [4] **W. Liu**, K. Wu, T. Liu, Y. Wang, K.-H. Yap, L.-P. Chau. “ByteNet: Rethinking Multimedia File Fragment Classification through Visual Perspectives.” *IEEE Transactions on Multimedia (TMM)*, 2024. [CCF-A, JCR Q1] [\[Representation Learning\]](#)
- [5] **W. Liu**, Y. Wang, K. Wu, K.-H. Yap, L.-P. Chau. “A Byte Sequence Is Worth an Image: CNN for File Fragment Classification Using Bit Shift and n-Gram Embeddings.” *IEEE International Conference on Artificial Intelligence Circuits and Systems (AICAS)*, 2023. [\[Representation Learning\]](#)
- [6] **W. Liu**, T. Liu, C. Cai, J. Gao, K. Wu, K.-H. Yap, C.-W. Lin. “AsTaSR: Adaptive Superpixel Token Aggregation for Lightweight Image Super-Resolution.” Submitted to *IEEE Transactions on Multimedia (TMM)*, 2026. [CCF-A, JCR-Q1] [\[Image Restoration\]](#)
- [7] **W. Liu**, D. Zhang, T. Liu, K. Wu, A. W.-K. Kong. “DriftAD: Visually-Guided Text Drift for Few-Shot Industrial Anomaly Detection.” Submitted to *ACM International Conference on Multimedia (ACM MM)*, 2026. [CCF-A] [\[Multimodal Vision\]](#)
- [8] T. Liu, K. Wu, Y. Wang, **W. Liu**, K.-H. Yap, L.-P. Chau. “Bitstream-Corrupted Video Recovery: A Novel Benchmark Dataset and Method.” *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. [CCF-A] [\[Image Restoration\]](#)
- [9] C. Cai, T. Liu, J. Gao, **W. Liu**, K. Wu, R. Wang, Y. Wang, S. C. Liew. “From Semantics, Scene to Instance-awareness: Distilling Foundation Model for Open-vocabulary Grounded Situation Recognition.” *ACM International Conference on Multimedia (ACM MM)*, 2025. [CCF-A] [\[Multimodal Vision\]](#)
- [10] R. Wang, S. Wang, R. Gong, C. Cai, J. Gao, W. Wang, **W. Liu**, K.-H. Yap. “MotionAnimate: Animate Human Images with Pose Motion for Vivid and Temporally Consistent Video Generation.” *Information Fusion*, 2025. [JCR Q1] [\[Multimodal Vision\]](#)
- [11] J. Gao, C. Cai, R. Wang, **W. Liu**, K.-H. Yap, K. Garg, B. S. Han. “CL-HOI: Cross-Level Human–Object Interaction Distillation from Multimodal Large Language Models.” *Knowledge-Based Systems (KBS)*, 2025. [JCR Q1] [\[Multimodal Vision\]](#)
- [12] **W. Liu**, W. Liu, M. Li, P. Chen, L. Yang, C. Xiao, Y. Ye. “Fine-Grained Task-Level Parallel and Low-Power H.264 Decoding in Multi-Core Systems.” *IEEE International Conference on Parallel and Distributed Systems (ICPADS)*, 2018. [CCF-C] [\[Efficient AI\]](#)
- [13] W. Liu, **W. Liu** (co-first), Y. Ye, Q. Lou, Y. Xie, L. Jiang. “HolyLight: A Nanophotonic Accelerator for Deep Learning in Data Centers.” *Design, Automation & Test in Europe Conference (DATE)*, 2019. [CCF-B] [\[Efficient AI\]](#)
- [14] Q. Lou, **W. Liu** (co-first), W. Liu, F. Guo, L. Jiang. “MindReading: An Ultra-Low-Power Photonic Accelerator for EEG-Based Human Intention Recognition.” *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2020. [CCF-C] [\[Efficient AI\]](#)